

CSE423 - Computer Graphics

Section 18 - Assignment: 1

Total Marks: 30

Deadline: 11:59 PM, 10th July 2026 | No late submissions will be accepted.

Assignment Tasks:

1. Explain the rules of the Digital Differential Analyzer (DDA) algorithm based on the slope of the line. Why does the original DDA algorithm fail when drawing a line from (7,2) to (2,5), and how does the "Revised DDA Algorithm" handle this scenario? [5]
2. Suppose a line segment starts at (120, 23) and ends at (423, 428). If we draw this problem using the DDA algorithm, determine the number of times y will be increased. How many pixels will be needed to draw this line? [5]

$$\frac{x}{7} - \frac{y}{12} = 5$$

3. Suppose a line segment has the equation above, and the starting point is at $y = 12$ and the ending point is at $y = 0$. Find the first 6 pixels using the Midpoint Line Algorithm. [5]
 4. Compare Vector and Raster displays. Which technology is better suited for displaying photorealistic images, and why? [5]
 5. A line has the equation $12x - 5y = 0$, starting at $y = 12$ and ending at $y = 0$. Using the Midpoint Line Drawing Algorithm and 8-way symmetry, determine the first 6 pixels. [5]
 6. A line segment has the starting point (20, -10) and the ending point (-30, 10). Using the Midpoint Line Algorithm and 8-way symmetry, determine the first 7 pixels. [5]
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Submission Guidelines:

1. You MUST write the answers on the paper. You can also use a Tab.
2. Make a cover page including your Assignment Number, Name, ID, Section, and Date.
3. Scan the pages and make a PDF. The file name should follow the format: StudentID_Section_Assignment_1 (eg 10301040_18_Assignment_1)
4. Upload the PDF to the [Google Form](#). Use G-suite Mail only.